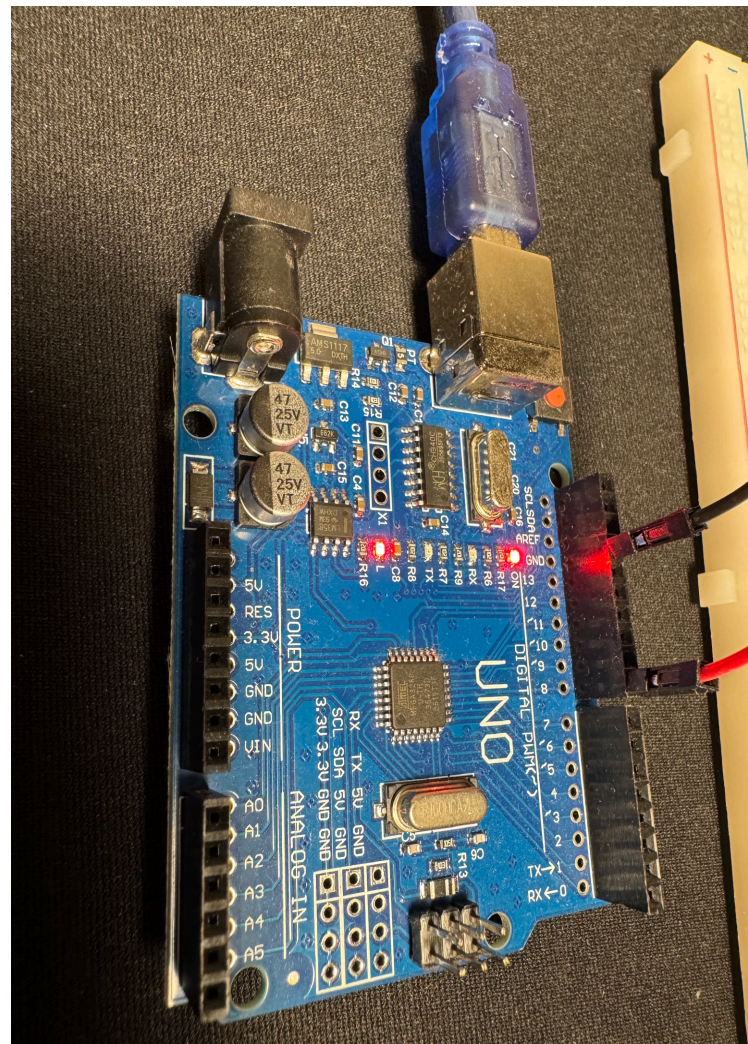
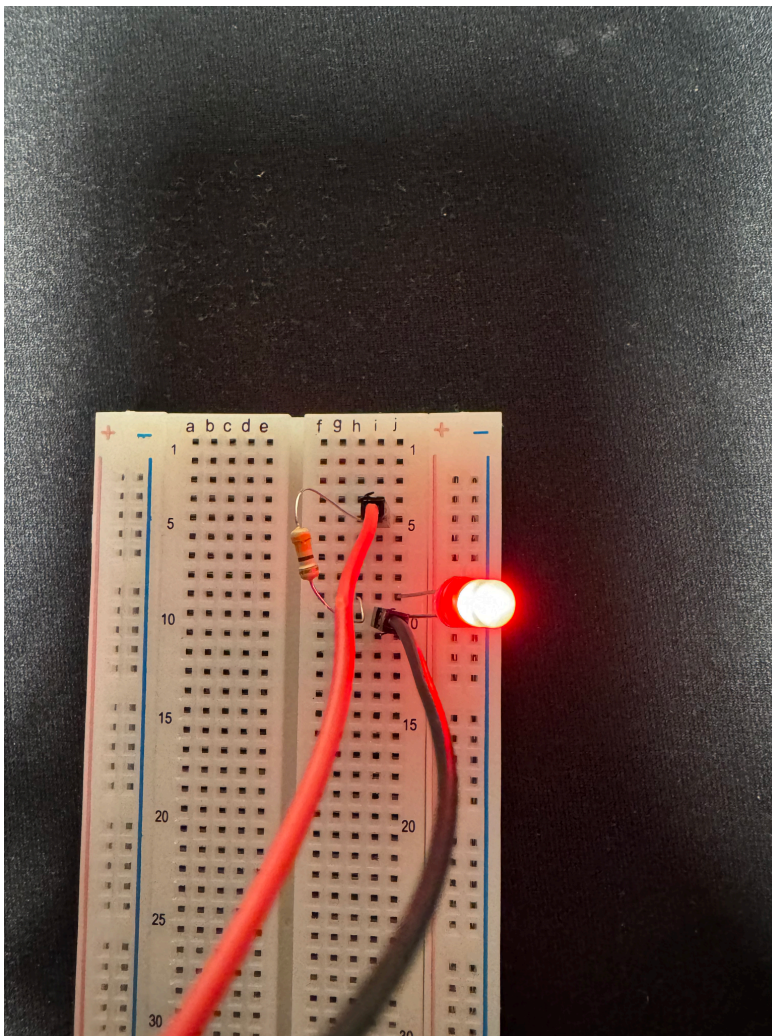
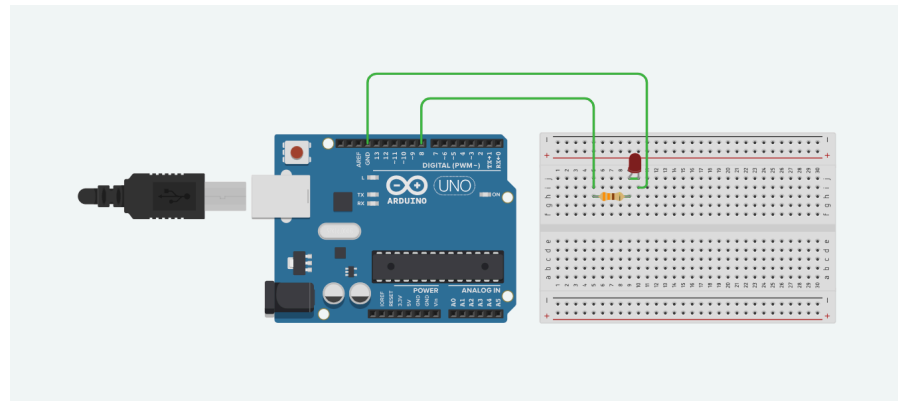


Blink



Circuit Components

- LED (light-emitting diode)

LEDs typically operate at around 15mA (milliampere). Applying 5V directly may damage the LED, so it must be protected with a resistor.

Ohm's Law: $V = I \times R$

$V \rightarrow$ Voltage (Volt (V))

$I \rightarrow$ Current (Ampere (A))

$R \rightarrow$ Resistance (Ohm (Ω))

- Resistor

LEDs cannot tolerate excessive current. Therefore, a 330 Ω resistor is used to protect the LED.

$$R = V / I \rightarrow 5 / 0.015 = 333.33\Omega$$

- Breadboard

A breadboard is a reusable plastic block filled with small holes used to build and test electronic circuits without soldering.

- Arduino

A printed circuit board that houses the microcontroller chip alongside extra supportive parts like power regulators, USB communication interfaces and easy-to-use pin headers.

- Jumper Wires

Jumper wires are short electrical wires with a pin or connector at each end, used to join two points in a circuit without soldering.

Circuit Connection

1. Connect the LED's short leg (cathode, -) directly to one of the Arduino's GND pins using a jumper wire.
2. Connect the LED's long leg (anode, +) to Arduino pin 8 through a 330 Ω resistor.

What Happens?

- When pin 8 is HIGH, it outputs about 5V and the LED turns on.
- When pin 8 is LOW, it outputs 0V and the LED turns off.
- `delay(500)` pauses the program for 500 milliseconds which is half a second.

Pin 8 $\rightarrow$ 330 Ω resistor $\rightarrow$ LED's long leg (anode) $\rightarrow$ LED $\rightarrow$ short leg (cathode) $\rightarrow$ GND

Code

```
#define red_led 8

void setup() {
    // put your setup code here, to run once:
    pinMode(red_led, OUTPUT); // We just say to Arduino that pin 8 is an
    output port.
}

void loop() {
    // put your main code here, to run repeatedly:
    digitalWrite(red_led, HIGH); // We send a HIGH signal (5V) to pin 8. So,
    the LED turns on.
    delay(500); // Wait for 500ms. In other words, Let 500ms pass while the
    LED is on.
    digitalWrite(red_led, LOW); // We send a LOW signal (0V) to pin 8. So,
    the LED turns off.
    delay(500); // Wait for 500ms. In other words, Let 500ms pass while the
    LED is off.
}
```